nature energy

Article

https://doi.org/10.1038/s41560-026-02007-8

Bifacial tunnel oxide passivating contacts for silicon and perovskite/silicon tandem solar cells with improved efficiency

Received: 15 August 2025

Accepted: 10 February 2026

Published online: 11 March 2026

Check for updates

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The power conversion efficiency (PCE) of conventional tunnel oxide passivating contact (TOPCon) solar cells is fundamentally constrained by front-side recombination losses in both contact and non-contact regions. Here we demonstrate full-size bifacial TOPCon solar cells incorporating patterned front n-type TOPCon fingers and a full-area rear p-type TOPCon emitter, achieving a certified PCE of 26.34%. The devices exhibit excellent damp-heat stability and negligible light-induced degradation and light-and-elevated-temperature-induced degradation. These advances arise from the engineering of the front n-type TOPCon and rear bilayer p-type TOPCon contacts, enabled through controlled polycrystalline silicon crystallinity, dopant concentration, tunnel oxide properties and optimized silver paste formulation. Integrating this high-performance bifacial TOPCon bottom cell with a wide-bandgap perovskite top cell yields monolithic perovskite/TOPCon tandems with a certified PCE of 32.73% and an open-circuit voltage of 1.961 V. This work provides a scalable and industry-compatible pathway to higher-efficiency TOPCon and perovskite/TOPCon tandem photovoltaics.

High-efficiency crystalline silicon (c-Si) solar cells rely on advanced passivating contacts[1,2], such as the silicon heterojunction (SHJ)[3,4] and tunnel oxide passivating contact (TOPCon)[5] technologies. SHJ devices currently define the practical efficiency ceiling, achieving record power conversion efficiencies (PCEs) of 26.8% and 27.8% for front and back contact (FBC) and interdigitated back contact configurations, respectively[6]. The record PCE of monolithic perovskite/c-Si tandem (34.85%) also relies on an SHJ bottom cell[7]. Despite the overwhelming commercial dominance of TOPCon technology with rapidly expanding global market share[8], its certified efficiencies, 26.66% for full-size FBC devices[9] and 31.7% for perovskite/TOPCon tandems[10], still lag behind their SHJ counterparts.

Conventional TOPCon solar cells employ n-type c-Si wafers with a boron-diffused p^+ emitter at the front and a full-area SiO_x/phosphorus-doped polycrystalline silicon (poly-Si) (SiO_x/poly-Si(n^+)) passivating contact at the rear. Recently, a certified PCE of 26.35% was

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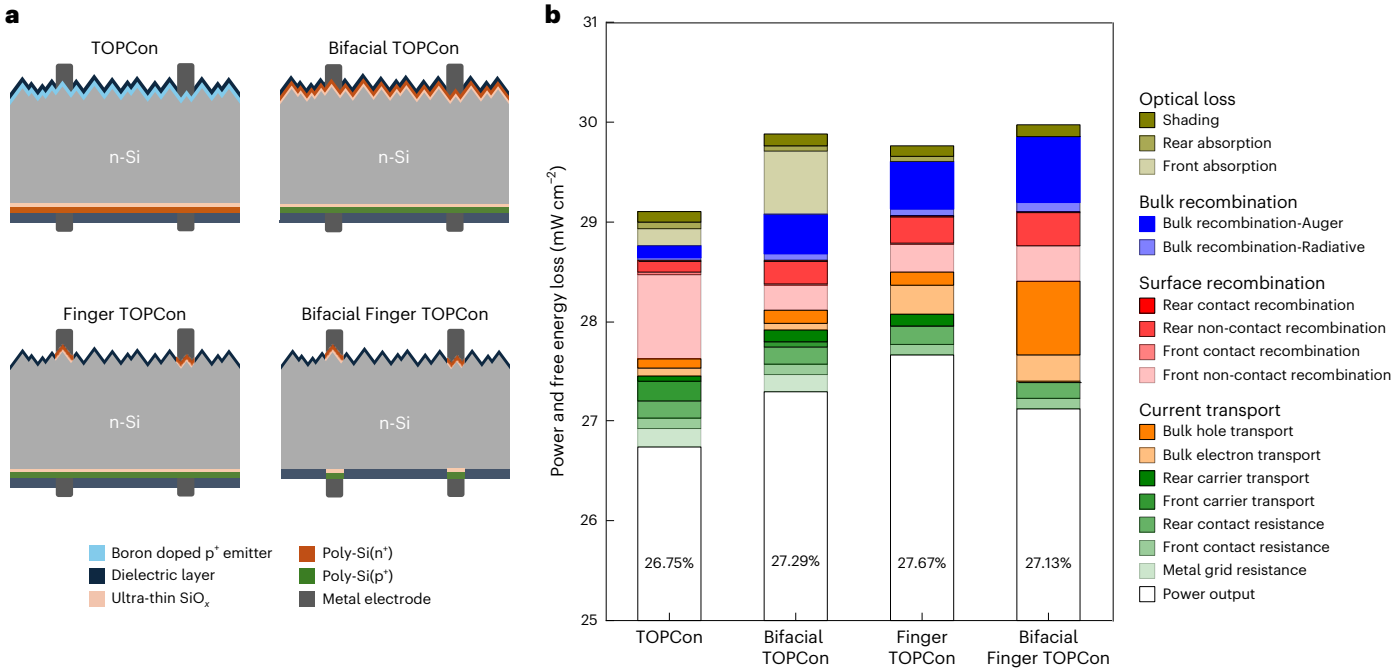

Fig. 1 | Simulated PCE potential of different TOPCon architectures. a, Schematic diagram of four TOPCon architectures, including TOPCon, Bifacial TOPCon, Finger TOPCon and Bifacial Finger TOPCon. **b**, PCE potential and corresponding free energy loss analysis of four TOPCon architectures.

demonstrated using a laterally modulated bilayer SiO_x/poly-Si(n^+) structure[11]. Further improvements, particularly high-sheet-resistance boron emitters and optimized grid designs, pushed efficiencies to 26.66%, driven mainly by enhanced front-side passivation and contact performance[9]. However, these advances do not overcome the intrinsic drawbacks associated with the boron-diffused p^+ emitter itself. Significant recombination losses arise from Auger recombination in the heavily doped p^+ emitter, Shockley–Read–Hall recombination at metal–silicon interfaces and bulk recombination within the c-Si absorber[12]. Theoretically, replacing the front p^+ emitter with a p-type TOPCon could raise the PCE limit to 28.7% by mitigating recombination losses, according to Brendel's calculation[13]. Yet, practical implementation faces a key trade-off: achieving both low dark recombination parameter (J_0) and low contact resistivity (ρ_c) during metallization by fire-through screen-printing pastes requires ≥100-nm-thick doped poly-Si, which induces substantial front-side parasitic absorption[12–17]. Patterned TOPCon structures have therefore been proposed to localize poly-Si only beneath metal fingers[18,19], mitigating optical losses while reducing recombination in non-contact regions[17,19–22]. Rear-junction architecture designs provide an additional benefit to this concept by enabling lateral carrier transport through the wafer to the front electrodes[23]. Simulations predict PCEs exceeding 26% for rear-junction devices incorporating localized front SiO_x/poly-Si(n^+) combined with full-area rear SiO_x/poly-Si(p^+)[24]. Nevertheless, experimentally realized PCEs remain far below expectations, typically around 22% (ref. 22), primarily because of the challenges in fabricating high-quality patterned SiO_x/poly-Si(n^+) on textured surfaces and firing-compatible SiO_x/poly-Si(p^+) compatible with industrial robustness[19,22,23].

In this work, we address both longstanding challenges and establish a high-efficiency bifacial TOPCon architecture featuring patterned front n-type TOPCon fingers and a full-area rear p-type TOPCon contact (Finger TOPCon). By optimizing SiO_x/poly-Si(n^+) on textured surfaces and engineering bilayer SiO_x/poly-Si(p^+) contacts on planar surfaces, we simultaneously achieve very low J_0 and ρ_c suitable for industrial firing processes. As a result, full-size Finger TOPCon solar cells deliver a certified PCE of 26.34%, with a high V_{OC} of 743.2 mV. Furthermore, integrating this bifacial TOPCon bottom cell into monolithic perovskite/TOPCon tandems yields a certified efficiency of 32.73%.

PCE potential of different TOPCon architectures

The practical PCE potential of four different TOPCon architectures was simulated using Quokka 3[25], with key achievable parameters listed in Supplementary Table 1[6,24,26–29] and detailed in Supplementary Note 1 and Supplementary Figs. 1–6. As shown in Fig. 1a, all architectures are n-type c-Si solar cells featuring distinct configurations: the conventional TOPCon employs a full-area boron-diffused p^+ front emitter with full-area rear SiO_x/poly-Si(n^+); bifacial TOPCon utilizes full-area front SiO_x/poly-Si(n^+) and rear SiO_x/poly-Si(p^+); finger TOPCon introduces localized front SiO_x/poly-Si(n^+) fingers while retaining full-area rear SiO_x/poly-Si(p^+); in contrast, the Bifacial Finger TOPCon implements localized TOPCon contacts on both front and rear surfaces. Note that the latter three configurations adopt rear-junction designs, motivated by the challenge of achieving highly passivated p-type TOPCon on textured surfaces and the advantage of lateral electron transport in n-Si substrates.

Figure 1b presents the calculated PCE limits and power loss analysis. The conventional TOPCon shows the lowest PCE potential at 26.75%, primarily limited by front non-contact recombination at the p^+ emitter. Transitioning to bifacial architectures, either full-area or localized, mitigates these recombination pathways and increases the PCE potential beyond 27%. The Bifacial TOPCon displays a PCE potential of 27.29%, which was mainly limited by the front parasitic absorption loss of the poly-Si(n^+) even with a thickness of 10 nm. The Finger TOPCon achieves the highest PCE potential at 27.67%, enabled by significantly reduced parasitic absorption due to the patterned poly-Si(n^+). In comparison, the PCE of Bifacial Finger TOPCon drops to 27.13% due to reduced hole collection efficiency and increased bulk recombination, and this limitation persists irrespective of the rear finger pitch (Supplementary Fig. 7). It should also be noted that bifaciality is not considered in these simulations but is relevant for module-level performance in practical applications. Across all architectures, improvements in surface passivation shift the dominant loss mechanism toward intrinsic Auger recombination, highlighting the fundamental limitations of highly passivated

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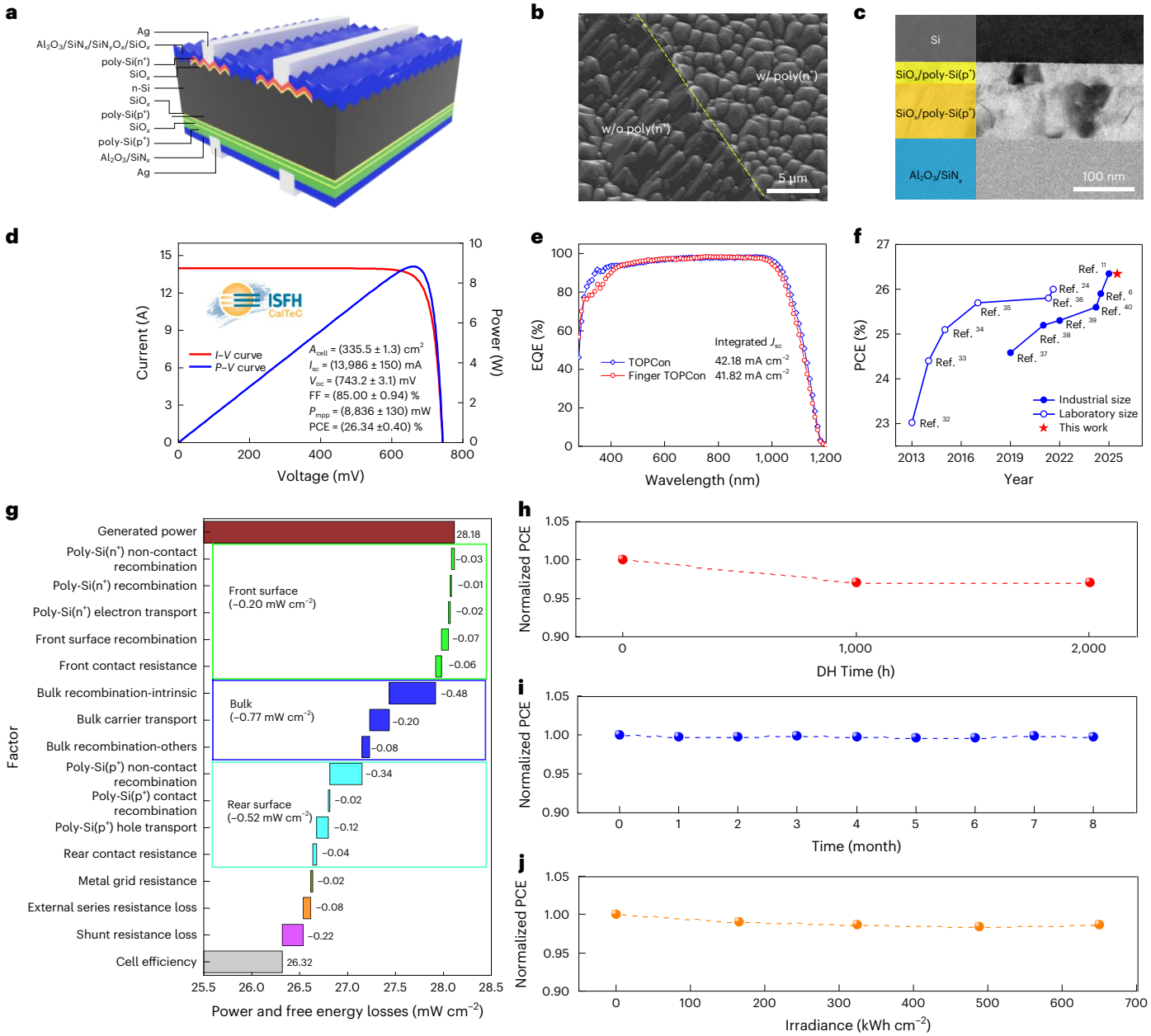

Fig. 2 | Performance of Finger TOPCon solar cells. a, Schematic diagram of Finger TOPCon solar cell. **b**,**c**, SEM images of the patterned front SiO_x/poly-Si(n^+) (**b**) and rear bilayer SiO_x/poly-Si(p^+) passivating contact (**c**). **d**,**e**, Certified current–voltage (*I*–*V*) and power–voltage (*P*–*V*) curve (**d**) and EQE of Finger TOPCon solar cell (**e**). f, PCE evolution for FBC TOPCon solar cells[6,11,12,23,48–55] over the years. **g**, Power and free energy loss analysis of Finger TOPCon solar cell. **h**–**j**, Stability of single-cell Finger TOPCon module encapsulated with glass/glass under accelerated DH (85 ± 2 °C, 85% relative humidity, in air, in dark) (**h**), LID (50 ± 10 °C, in air, 1,000 W m^{-2} from LED-based solar simulator for 60 h per month) (**i**) and LeTID (75 ± 2 °C, in air, varying illumination intensity for 324 h from LED-based solar simulator) (**j**). The absolute starting PCE values in **h**, **i** and **j** are 25.82%, 25.86% and 25.75%, respectively. Credit: **d**, ISFH.

c-Si solar cells. Consequently, Finger TOPCon emerges as the most promising architecture among the four configurations.

Finger TOPCon solar cell performance

Guided by simulation results, we fabricated industrial-size Finger TOPCon solar cells featuring high-quality localized SiO_x/poly-Si(n^+) contacts on textured front-side and full-area SiO_x/poly-Si(p^+) contacts on the polished rear side, as illustrated in Fig. 2a. The front SiO_x/poly-Si(n^+) fingers (~210 μm width) were fabricated via UV laser oxidation patterning followed by alkaline etching (Fig. 2b), and subsequently capped with $Al_2O_3/SiN_x/SiN_yO_x/SiO_x$ passivation and anti-reflection stack. The 210 μm finger width reflects the current limitations of laser patterning precision and optical alignment of the screen-printing process. The rear side incorporates a full-area bilayer TOPCon structure (SiO_x/poly-Si(p^+)/SiO_x/poly-Si(p^+)) capped with Al_2O_3/SiN_x (Fig. 2c). Front and rear finger electrodes were formed using fire-through silver paste.

The total-area Finger TOPCon solar cell (335.5 cm^2) achieves a certified PCE of 26.34%, featuring a high V_{oc} of 743.2 mV, FF of 85.0% and J_{sc} of 41.69 mA cm^{-2} (Fig. 2d). This significantly outperforms conventional TOPCon with laser-doped selective emitter (25.4%), resulting from a lower V_{oc} of 719.1 mV, a lower FF of 83.66% and a higher J_{sc} of 42.24 mA cm^{-2} (ref. 30). The 24.1 mV V_{oc} gain and 1.34%-absolute FF improvement confirm reduced front recombination and efficient lateral carrier transport via wafer bulk, aligning with simulations. The large absolute J_{sc} loss of 0.55 mA cm^{-2} primarily stems from the parasitic absorption of 210 μm wide front poly-Si(n^+) fingers

(Supplementary Fig. 8), matching well with the simulated loss of 0.57 mA cm^{-2} (Supplementary Fig. 9). Further optimization, particularly narrowing and thinning the poly-Si(n$^+$), is expected to reduce parasitic absorption in the non-contact regions while preserving carrier selectivity at the metal-contact interface. The integrated J_{sc} of 41.81 mA cm^{-2} derived from external quantum efficiency (EQE) (Fig. 2e) validates the measured value. As expected, the EQE reduction at short (300–400 nm) and long wavelengths (1,000–1,200 nm) mirrors the observed J_{sc} loss. The impressive PCE (26.34%) and V_{oc} (743.2 mV) are on par with the highly optimized TOPCon devices reported by Wang et al.[11] (Fig. 2f and Supplementary Fig. 10), demonstrating the high potential of Finger TOPCon.

We further analyse the power and free energy loss of the 26.34%-efficient Finger TOPCon device using Quokka simulation (Fig. 2g). Experimentally measured key parameters, including J_0 at non-contacted regions and ρ_c, are listed in Supplementary Table 2. The simulated photovoltaic parameters (V_{oc} 743.2 mV, J_{sc} 41.69 mA.cm^{-2}, FF 85.04%, η 26.32%) exhibits excellent agreement with the certified results in Fig. 2d, validating the accuracy of loss analysis. Front-side losses are minimal (0.20 mW cm^{-2}), demonstrating the excellent passivation and carrier selectivity provided by both the patterned SiO_x/poly-Si(n$^+$) and the $Al_2O_3/SiN_x/SiN_yO_x/SiO_x$ passivation stack. The second-largest loss (0.52 mW cm^{-2}) originates from contact resistance and recombination in the full-area bilayer SiO_x/poly-Si(p$^+$) rear contact, indicating that this component remains the primary bottleneck. The dominant loss overall (0.77 mW cm^{-2}) arises from bulk carrier transport and recombination, suggesting that further improvements would benefit most from higher-quality wafers.

The encapsulated single-cell Finger TOPCon module demonstrates exceptional long-term stability under damp heat, illuminated and irradiated conditions. Under accelerated damp-heat (DH) testing (85 ± 2 °C, 85% relative humidity), normalized PCE degrades only 2.93% after 1,000 h and 2.97% after 2,000 h (Fig. 2h). The module exhibits outstanding resistance to light-induced degradation (LID), retaining 99.8% of initial PCE following 8 months of cyclic illumination exposure (1,000 W m^{-2} for 60 h per month; Fig. 2i). Furthermore, they show minimal degradation during light-and-elevated-temperature-induced degradation (LeTID) testing, with <1.5% PCE loss after 324 h at 75 °C under irradiance of 165–650 kW h m^{-2} (Fig. 2j). In addition, the module also demonstrates high UV stability, with <0.9% degradation after 240 h of UV exposure at 250 W m^{-2} (60 °C, 40% humidity), as shown in Supplementary Fig. 11.

Patterned SiO_x/poly-Si(n$^+$) on textured front side

To achieve high-quality patterned SiO_x/poly-Si(n$^+$) on textured front surfaces, we optimized texture morphology, crystallinity and phosphorus doping concentration in poly-Si(n$^+$). Conventional TOPCon structures suffer from poor passivation on sharp upright pyramids (Supplementary Fig. 12) due to high defect densities, non-uniform SiO_x/poly-Si(n$^+$) deposition and susceptibility to silver paste corrosion at pyramid peaks[31,32]. To address this, we introduce round pyramids[31,33] by implementing an industrially compatible solution process using ozone and hydrofluoric acid (Supplementary Table 3). As shown in Fig. 3a, the rounded-pyramid peaks significantly reduce average surface roughness, which facilitates uniform SiO_x/poly-Si(n$^+$) deposition while preventing Ag paste corrosion during metallization. This markedly improves passivation quality in both contact and non-contact regions (Supplementary Fig. 13). Crucially, pyramid rounding induces negligible reflection loss when coated with the $Al_2O_3/SiN_x/SiN_yO_x/SiO_x$ passivation/anti-reflection stack (Supplementary Fig. 14), which also provides excellent surface passivation in front non-contact regions (J_0: 0.5 fA cm^{-2}; implied V_{oc}: 747 mV) (Supplementary Figs. 15 and 16).

To enhance poly-Si(n$^+$) crystallinity on textured surfaces, we developed a gradient thermal field (GTF) within the low-pressure chemical vapour deposition (LPCVD) system by incorporating four circular temperature-control elements (Supplementary Fig. 17). The conventional single thermal field can introduce temperature non-uniformity within the LPCVD tube, leading to premature gas-phase reactions and reduced film crystallinity. In contrast, the GTF greatly improves horizontal and vertical temperature uniformity, thereby enabling the formation of highly oriented, low-defect poly-Si films. Compared with conventional thermal-field deposition (Fig. 3b), poly-Si films under GTF exhibit higher crystallinity with substantially enlarged grains, as evidenced by top-view and cross-sectional scanning electron microscope (SEM) imaging (Figs. 3c,d). Raman spectroscopy further confirms this improvement, showing a narrowed full width at half maximum (FWHM) of 8.4 cm^{-1} versus 11.9 cm^{-1} for conventional processing (Fig. 3e). The 1.9 cm^{-1} peak shift toward higher wavenumbers indicates reduced tensile strain and elongated Si–Si bonds with lower internal energy[34]. Consequently, the best iV_{oc}, implied FF (iFF) and J_0 of SiO_x/poly-Si(n$^+$) under GTF are significantly improved from 718 mV, 85.4% and 10.1 fA cm^{-2} to 742 mV, 86.5% and 1.0 fA cm^{-2} (Fig. 3f and Supplementary Fig. 16), primarily due to the enhanced doping efficiency of highly-crystallized poly-Si.

We next investigated the effect of ex situ diffused phosphorus doping concentrations on the passivation performance of SiO_x/poly-Si(n$^+$) on the textured surface. By tuning the ex situ diffusion parameters (for example, temperature and time), the phosphorus doping concentration in the poly-Si varied from 1.5×10^{20} cm^{-3} to 4.3×10^{20} cm^{-3} (Fig. 3g), as measured by the electrochemical capacitance–voltage technique (ECV). The doping concentration cliff defines the SiO_x/poly-Si(n$^+$) interface, indicating ~120 nm poly-Si thickness. As concentration increases from 1.5×10^{20} cm^{-3} to 3.3×10^{20} cm^{-3}, the average J_0 decreases from 7.2 fA cm^{-2} to 1.0 fA cm^{-2} while the average iV_{oc} increases from 726 mV to 740 mV (Fig. 3h). The improved passivation observed with increasing phosphorus doping concentration can be attributed to the enhanced field-effect passivation, driven by higher doping efficiency enabled by improved poly-Si(n$^+$) crystallinity under the combined influence of the gradient thermal field and rounded-pyramid texture. When the doping level exceeds ~3.3×10^{20} cm^{-3}, J_0 increases due to defect generation and possible disruption of the SiO_x interlayer during thermal processing. In contrast, ρ_c decreases steadily with increasing doping concentration because of the improved electrical properties of the poly-Si(n$^+$) layer (Fig. 3i). A very low ρ_c of 0.61 mΩ cm^2 is achieved at the optimal doping concentration of 3.3×10^{20} cm^{-3}, enabling efficient electron collection with localized contact architecture. With all optimizations combined, the optimized SiO_x/poly-Si(n$^+$) passivated contact and $Al_2O_3/SiN_x/SiN_yO_x/SiO_x$ stack achieve a remarkably low total J_0 of 0.6 fA cm^{-2} on the textured front surface, partly enabling the high V_{OC} of 743.2 mV.

Rear bilayer SiO_x/poly-Si(p$^+$) passivating contact

To advance rear hole-selective passivating contact performance, we engineered a bilayer SiO_x/poly-Si(p$^+$) structure on polished c-Si surfaces, fabricated using a standard industrial alkali polishing process (Supplementary Fig. 18) and developed a specialized glass frit component for silver paste. As resolved by high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) and energy-dispersive X-ray spectroscopy (EDX) (Fig. 4a), the architecture comprises SiO_x/poly-Si(p$^+$)/SiO_x/poly-Si(p$^+$) with poly-Si(p$^+$) thicknesses of 36 nm and 90 nm. A critical in situ oxidized SiO_x interlayer (~ 1 nm) between poly-Si(p$^+$) layers aims to mitigate Ag crystallite spiking during subsequent firing. The bilayer structure is capped with Al_2O_3/SiN_x stack for hydrogenation to enhance passivation.

We systematically optimized the bilayer SiO_x/poly-Si(p$^+$) by investigating tunnel oxide growth methods, thickness, crystallinity and boron doping of poly-Si. Among all oxidation routes, the in situ thermal oxidation at 650 °C in LPCVD achieves superior performance (average iV_{OC}: 745.4 mV; Supplementary Fig. 19), outperforming wet oxidation (HNO_3) and plasma-assisted N_2O oxidation. This can be ascribed to the denser and near-stoichiometric characteristic of thermally grown

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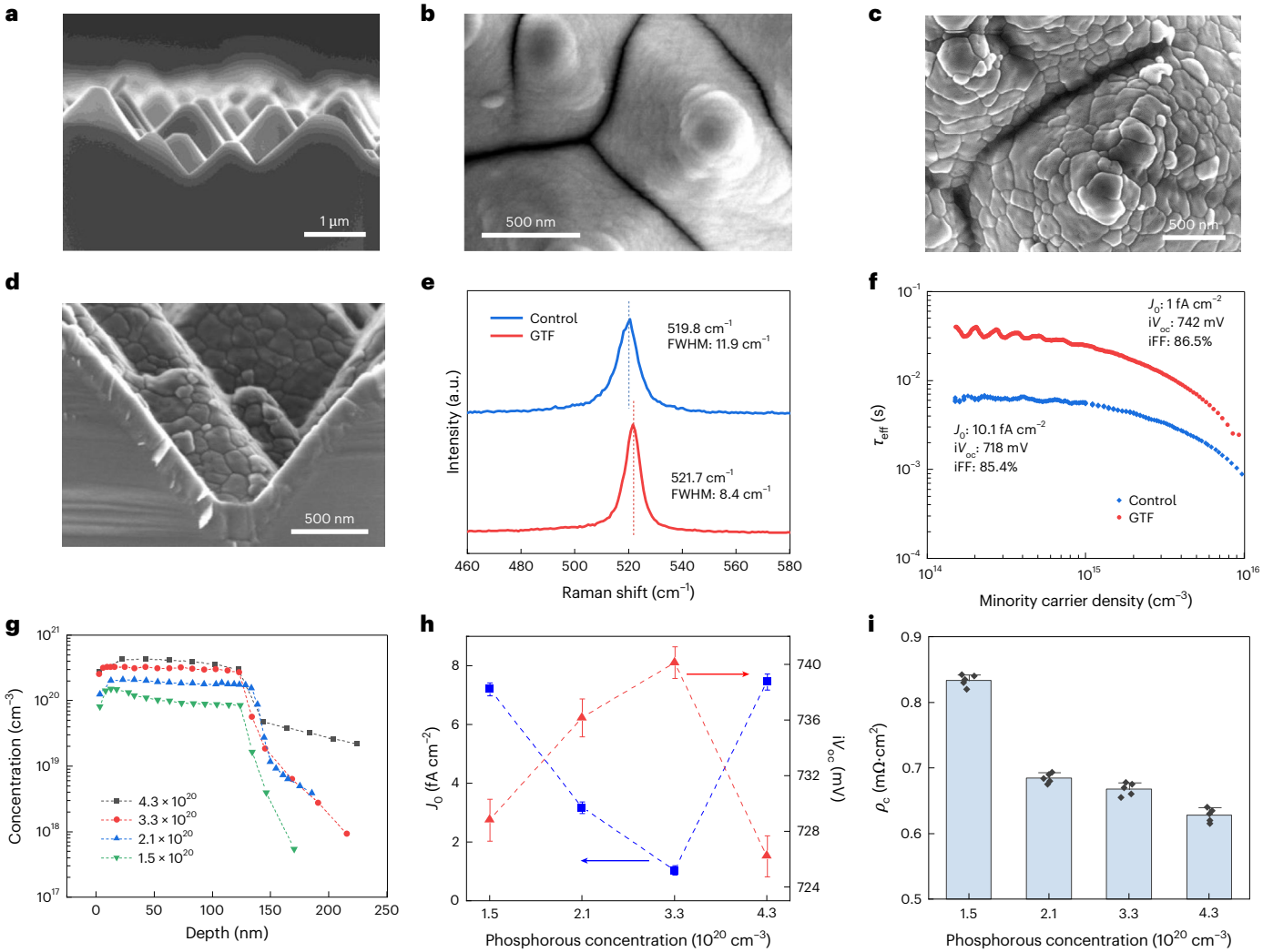

Fig. 3 | SiO_x/poly-Si(n$^+$) passivating contact on textured front surface. **a**, Cross-sectional SEM images of round pyramids. **b**,**c**, Top-view SEM images of poly-Si under conventional thermal field (**b**) and GTF (**c**). **d**, Cross-sectional SEM image of poly-Si under GTF. **e**, Raman spectra of poly-Si with and without GTF. **f**, Injection level dependent effective lifetime of textured n-Si passivated by SiO_x/poly-Si(n$^+$) with and without GTF. **g**, Phosphorous doping profile measured by ECV. **h**, J_0 and iV_{oc} of textured n-Si passivated by the SiO_x/poly-Si(n$^+$) as a function of phosphorous doping concentration. The error bars on mean values are standard deviations from five samples. **i**, ρ_c of SiO_x/poly-Si(n$^+$) on textured n-Si as a function of phosphorus doping concentration. The bar charts and error bars are mean value and standard deviations from five samples, respectively.

SiO_x, which prevents boron from accumulating in the SiO_x layer[35]. Crucially, high-quality SiO_x is more critical for SiO_x/poly-Si(p$^+$) than SiO_x/poly-Si(n$^+$), because boron tends to enrich in the SiO_x layer due to its smaller segregation coefficient (0.1 ~ 0.3) than phosphorus (~ 10)[26]. Figure 4b shows the iV_{OC} of n-Si passivated by bilayer SiO_x/poly-Si(p$^+$) as a function of SiO_x thickness. The iV_{OC} increases from 733 mV to 745.4 mV with increasing SiO_x thickness from 1.4 nm to 1.8 nm, as illustrated from high-resolution transmission electron microscopy images in Supplementary Fig. 20. Beyond this optimal thickness, iV_{OC} drops to 741 mV when SiO_x thickness reaches 2 nm. This is due to diminished field-effect passivation from reduced boron diffusion into the silicon surface. The SiO_x exhibits excellent spatial uniformity and reproducibility across wafers, tubes, and chambers (Supplementary Fig. 21), ensuring the high and consistent passivation performance of the developed contacts.

To examine how poly-Si(p$^+$) crystallinity affects bilayer SiO_x/poly-Si(p$^+$) performance, we introduced a pre-annealing treatment in N_2 atmosphere before boron diffusion. As-deposited silicon film exhibits amorphous, and its crystallinity was tuned by pre-annealing at 850 °C, 950 °C and 1,050 °C, respectively, as proven by the selected area electron diffraction and Raman spectrum (Supplementary Figs. 22 and 23). Pre-annealing at 1,050 °C yields highly crystalline poly-Si (FWHM = 3.7 cm^{-1}), and the superior crystallinity (FWHM: 4.9 cm^{-1}) is maintained after post-diffusion, outperforming the control sample (FWHM: 5.5 cm^{-1}) (Fig. 4c). Correspondingly, the average iV_{OC} of n-Si passivated by bilayer SiO_x/poly-Si(p$^+$) dramatically increases from 698.9 mV to 745.7 mV (Fig. 4d) due to high crystallinity of ~98% after introducing pre-annealing treatment[36]. This is related to improved field-effect passivation due to enhanced doping efficiency in highly crystalline poly-Si (Supplementary Fig. 24). Moreover, the high-temperature pre-annealing treatment promotes stoichiometry of the tunnel oxide and generates moderate pinholes in the tunnel oxide[37], which improves the chemical passivation and reduces the tunnel oxide damage by boron atoms with an alternative pinhole transport path. Boron doping concentration also critically influences the passivation of bilayer SiO_x/poly-Si(p$^+$), as shown in Fig. 4e. With increasing boron doping concentrations from 4×10^{19} cm^{-3} to 1×10^{20} cm^{-3} (ECV boron doping profile in Supplementary Fig. 25), the average iV_{OC} increases from 725 mV to 745.7 mV. Beyond 1×10^{20} cm^{-3}, Auger recombination and boron-related defects degraded performance[38,39]. Note that the bilayer

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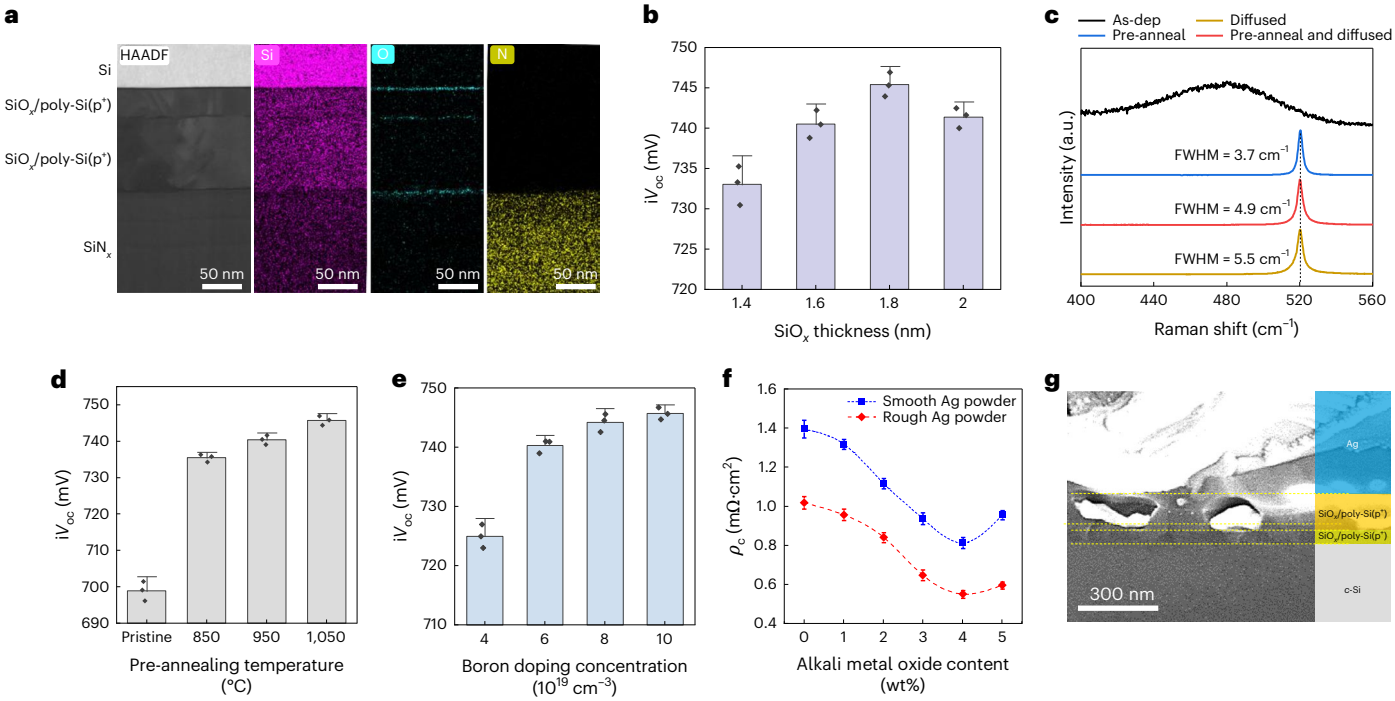

Fig. 4 | Rear bilayer SiO_x/poly-Si(p^+) passivating contact. a, HAADF and EDX mapping image of bilayer SiO_x/poly-Si(p^+) structure. **b**, iV_{oc} of the planar n-Si passivated by bilayer SiO_x/poly(p^+) as a function of thermally grown SiO_x thickness. **c**, Raman spectra of boron-diffused poly-Si films with and without pre-annealing treatment. **d**,**e**, iV_{oc} of planar n-Si passivated by bilayer SiO_x/poly-Si(p^+) as a function of crystallinity of poly-Si(p^+) (**d**) and boron doping concentration (**e**). The bar charts and error bars in **b**, **d** and **e** are mean value and standard deviations from three samples, respectively. **f**, ρ_c of bilayer SiO_x/poly-Si(p^+) on n-Si as a function of alkali metal oxide content in Ag powder. The error bars on mean values in **f** are standard deviations from six samples. **g**, Cross-sectional SEM image of the contact interface between bilayer SiO_x/poly-Si(p^+) and Ag electrode after firing.

SiO_x/poly-Si(p^+) structure yields higher passivation performance than the control sample, in which the poly-Si (126 nm) was deposited continuously. The bilayer configuration improves iV_{OC}, J_0 and iFF from 734 mV, 6 fA cm^{-2} and 86.1% to 747 mV, 2.2 fA cm^{-2} and 86.4%, respectively (Supplementary Fig. 26). The reason can be ascribed to that the bilayer configuration can effectively prevent the in-diffusion of boron and mitigate the detrimental effects of boron segregation-induced damage on the interfacial oxides[26]. Ultimately, the optimized bilayer SiO_x/poly-Si(p^+) achieves excellent passivation performance with a high iV_{oc} of 747 mV and a low J_0 of 2.2 fA cm^{-2}.

To enhance the contact performance of the bilayer SiO_x/poly-Si(p^+), we engineered a modified Ag paste. Conventional screen-printed fire-through metallization performs poorly for SiO_x/poly-Si(p^+) versus SiO_x/poly-Si(n^+)[40]. Our solution incorporated alkali metal oxides (sodium oxide and potassium oxide) into the glass frit, reducing its melting point and transforming the glass network from chain-like to island structures. This improved flow characteristics and progressively reduces ρ_c, reaching a minimum of 0.82 mΩ cm^2 at 4 wt% alkali content (Fig. 4f). Beyond this level, ρ_c increases probably due to the enhanced corrosion damage of poly-Si(p^+) after firing. Introducing rough Ag powder further reduces ρ_c to 0.55 mΩ cm^2 (Supplementary Fig. 27 and Fig. 4f), facilitated by increased contact sites from higher surface area[41,42]. The modified paste also improves Ag-finger aspect ratios, yielding 8-µm height and 20-µm width (Supplementary Fig. 28). Critically, the bilayer structure effectively suppresses Ag crystallite spiking during firing. Cross-sectional SEM (Fig. 4g) shows that Ag crystallites penetrate only the outer poly-Si(p^+) layer and are halted by the thin SiO_x interlayer. HAADF and energy-dispersive spectroscopy (EDS) images (Supplementary Fig. 29) confirm that large Ag particles are predominantly blocked outside the inner SiO_x/poly-Si layer, while only small Ag particles reach the inner poly-Si without contacting the c-Si substrate. In contrast, extensive Ag penetration into the silicon bulk is observed for the single SiO_x/poly-Si configuration. The suppression of spiking probably results from the electron deficiency of the SiO_x interlayer, which hinders the reduction of Ag^+ during cooling and disrupts vertical grain alignment, preventing the formation of continuous penetration pathways[43,44]. Quokka simulations estimate a low rear-contact J_0 of 5 fA cm^{-2} (Supplementary Table 2), contributing significantly to the high V_{OC} of the Finger TOPCon device.

Perovskite/TOPCon tandem solar cell

The efficiency of perovskite/TOPCon tandem solar cells is mainly limited by both V_{OC} and J_{SC} (Supplementary Table 4). The inferior V_{OC} originates from the intrinsically lower V_{OC} of conventional TOPCon bottom cells, whereas the commonly used planar configuration limits J_{SC} because of poor light trapping. To address these limitations, we fabricated bifacial TOPCon solar cells with full-area SiO_x/poly(n^+) on textured front and bilayer SiO_x/poly(p^+) on the rear, achieving a high PCE of 24% with V_{OC} 741.3 mV, J_{SC} of 38.2 mA cm^{-2} and FF 84.8% (Supplementary Fig. 30). Leveraging the high V_{OC} and textured front surface of this bifacial TOPCon, we fabricated monolithic perovskite/TOPCon tandem solar cells (Fig. 5a). The tandem incorporates a wide-bandgap perovskite top cell (1.68 eV) with zinc-doped indium oxide (IZO) recombination layer, classical nickel oxide/4-(3,6-dimethyl-9H-carbazol-9-yl) butyl]-phosphonic acid (NiO_x/Me-4PACz) hole transport layers and fullerene/tin oxide (C_{60}/SnO_2) electron transport/buffer layers combined with IZO front electrode at the bottom and front side, respectively. Figure 5b shows the illuminated J–V curve of the champion tandem device (1 cm^2), achieving a certified PCE of 32.73% with negligible hysteresis (Supplementary Fig. 31). Stabilized power output under maximum power point (MPP) tracking confirms a high PCE of 32.3% (Fig. 5c). The tandem exhibits a high V_{OC} of 1.961 V, FF of 81.83% and J_{SC} of 20.40 mA cm^{-2}, demonstrating the high performance of as-developed bifacial TOPCon bottom cell. EQE measurement shows

https://doi.org/10.1038/s41560-026-02007-8

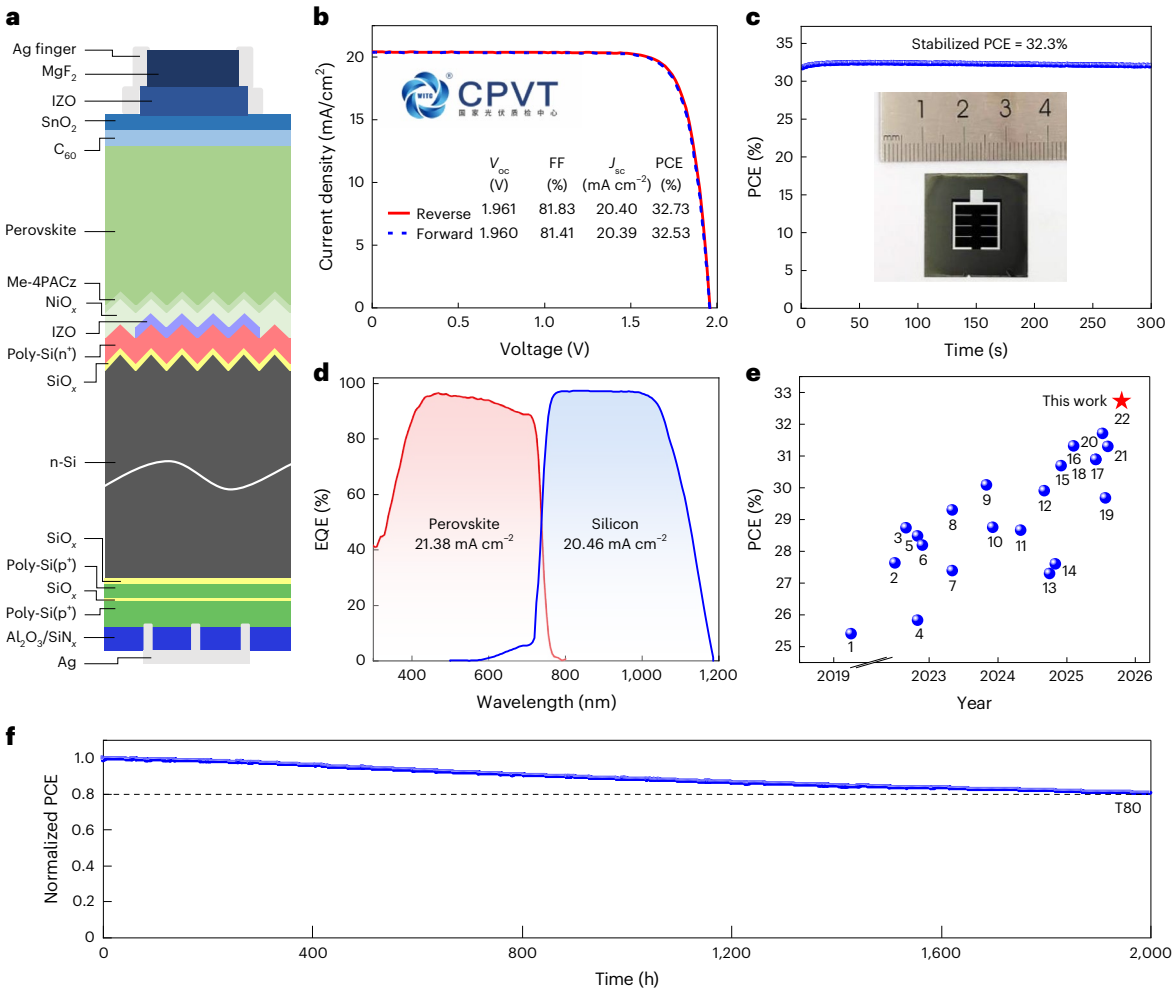

Fig. 5 | Performance of monolithic perovskite/TOPCon tandem solar cells. **a**, Schematic of the perovskite/TOPCon tandem. **b**, Certified *J*–*V* curves. **c**,**d**, Stabilized power output (**c**) and EQE of the champion device (**d**). The inset in **c** shows the photos of perovskite/TOPCon tandem solar cell. **e**, PCE evolution of perovskite/TOPCon tandems (all photovoltaic parameters are provided in Supplementary Table 4). **f**, Long-term MPP tracking of the encapsulated tandem under continuous illumination (AM 1.5 G, 1,000 W m^{-2}, LED-based solar simulator) at room temperature in nitrogen atmosphere. The absolute starting PCE value is 30.58%. Credit: **b**, CPVT.

21.38 and 20.46 mA cm^{-2} current outputs from the perovskite top cell and TOPCon bottom cell (Fig. 5d), respectively, which is in line with the certified J_{SC} of 20.4 mA cm^{-2}. Nevertheless, the J_{SC} of the tandem device remains limited by significant parasitic absorption in the poly-Si layers of the TOPCon bottom cell, as evidenced by optical simulations using SunSolve Power, with 0.97 mA cm^{-2} and 0.49 mA cm^{-2} loss for the front and rear poly-Si, respectively (Supplementary Table 5). According to our previous work[45], the perovskite top cell exhibited a V_{OC} of approximately 1.24 V, indicating that the bifacial TOPCon bottom cell contributed 0.72 V to the tandem V_{OC}. The resulting V_{OC} of 1.961 V is higher than previously reported values for perovskite/TOPCon tandem solar cells and comparable to the perovskite/SHJ counterpart[45–47]. The certified 32.73% efficiency also outperforms existing published results for monolithic perovskite/TOPCon tandem solar cells (Fig. 5e and Supplementary Table 4). Long-term stability was assessed by MPP tracking of encapsulated tandems under continuous 1-sun illumination at room temperature in a nitrogen atmosphere. The device maintains 80% of its initial PCE after 2,000 h (Fig. 5f), demonstrating exceptional operational stability. Although challenges remain for large-area uniformity and long-term stability of the perovskite top cells, rapid academic and industrial progress in scalable deposition techniques and interface engineering is paving the way towards commercially viable perovskite/TOPCon tandem technologies. Importantly, the bifacial TOPCon bottom cell demonstrated here is fully compatible with existing industrial production lines, facilitating future industrial adoption.

Conclusions

We demonstrate industrially compatible Finger TOPCon solar cells achieving a certified efficiency of 26.34%, enabled by patterned front SiO_x/poly-Si(n$^+$) fingers and a full-area rear bilayer SiO_x/poly-Si(p$^+$) passivating contact. This architecture overcomes intrinsic limitations of conventional full-area TOPCon by simultaneously suppressing front-side recombination through contact patterning and reducing rear-side recombination using a firing-stable bilayer SiO_x/poly-Si(p$^+$). The resulting V_{OC} of 743.2 mV and FF of 85.0% confirm the excellent carrier selectivity of both contacts. The devices exhibit strong reliability, with negligible degradation under DH, LID and LeTID testing. Looking forward, efficiencies approaching 27% in mass production appear attainable through additional reductions in parasitic absorption, by narrowing and thinning the patterned SiO_x/poly-Si(n$^+$) fingers and employing locally thinned SiO_x/poly-Si(p$^+$) layers at the rear. Finally, using the developed bifacial TOPCon bottom cell, we achieve a certified PCE of 32.73% in monolithic perovskite/TOPCon tandems, which also demonstrate outstanding long-term operational stability. These results establish a scalable and future-ready pathway for next-generation TOPCon and perovskite/TOPCon silicon tandem photovoltaics.

Methods

Finger TOPCon solar cells

Commercial n-type Cz silicon wafers (1 Ω.cm, 140 μm thick, 182.3 × 183.75 mm^2) were used for TOPCon solar cell fabrication. The process flow of Finger TOPCon solar cells is shown in Supplementary Fig. 32. After standard cleaning and subsequent alkali polishing, the rear-side bilayer SiO_x/poly-Si(p^+) structure was formed by first growing a 1.8-nm SiO_x layer, depositing a 36-nm amorphous silicon (a-Si) layer, then growing a 1-nm SiO_x interlayer via LPCVD, followed by deposition of a 90-nm a-Si layer at 550 °C. Both a-Si layers were deposited using identical LPCVD conditions. The stack was subsequently pre-annealed at 1,050 °C in N_2 atmosphere, followed by boron diffusion with a concentration of 1×10^{20} cm^{-3} in a tube furnace. Front-side processing included wrap-around etching, borosilicate glass removal in HF and texturing in alkaline solution to form random pyramids. The textured surface was conditioned using an HF/HCl/O_3 mixture for 100 s at 25 °C (Supplementary Table 3). A front SiO_x (1.2 nm) /poly-Si(n^+) (120 nm) bilayer with phosphorus doping concentration of 3.3×10^{20} cm^{-3} was deposited via LPCVD under GTF to enhance temperature uniformity, after which phosphorus silicate glass (PSG) was removed by HF etching. Localized front fingers (210 μm wide) were defined via 355-nm UV laser oxidation, followed by selective removal of non-contact poly-Si(n^+) in 9 vol% KOH at 40 °C. After RCA cleaning, a 3-nm Al_2O_3 passivation layer was deposited on both sides via ALD, and front SiN_x/SiN_yO_x/SiO_x multilayer (110 nm) and rear SiN_x layer (80 nm) were deposited by PECVD. Final metallization was achieved through screen printing of Ag pastes on both sides, followed by co-firing. The width of the front and rear screen-printed metal fingers is 16 μm and 20 μm, respectively.

TOPCon solar cells

Commercial n-Si wafers were textured in NaOH to form micrometre-scale pyramids after standard cleaning. Boron diffusion using BCl_3 gas formed the boron emitter, followed by the formation of front selective emitters using a laser beam width of 90–100 μm. After rear-side polishing, a SiO_x (1.2 nm)/poly-Si(n^+) (120 nm) stack was deposited on the rear surface by LPCVD. The PSG was formed during $POCl_3$ diffusion in an O_2/N_2 atmosphere, after which the a-Si layer was crystallized into poly-Si. PSG was removed in 5% HF for 5 min. Subsequently, ALD AlO_x (3 nm) and PECVD SiN_x (80 nm) were deposited on both sides. Finally, bifacial metal grid electrodes were fabricated by screen printing followed by a standard firing process.

Perovskite/TOPCon tandem solar cells

The perovskite/TOPCon tandem solar cells begin with Bifacial TOPCon bottom cells with full-area front SiO_x/poly-Si(n^+) on textured surface and bilayer SiO_x/poly-Si(p^+) on planar surface. The rear metal electrode was fabricated by screen-printed Ag paste. The fabrication process and parameters are the same as the Finger TOPCon solar cell, without the UV laser patterning step. The doping concentration of the rear poly-Si(p^+) and front poly-Si(n^+) layers is 1×10^{20} cm^{-3} and 3.3×10^{20} cm^{-3}, respectively. The active and inactive regions of bilayer SiO_x/poly-Si(p^+) are separated by etching. Note that the Al_2O_3 passivation layer was deposited and subsequent firing was performed to hydrogenate the front SiO_x/poly-Si(n^+) layer. The rear-side processing followed the same procedure as for single-junction Finger TOPCon solar cells. After the surface hydroxylation treatment by UV/O_3 cleaner (UVO5, CIF (Beijing) Tech Co., Ltd), a 5-nm IZO recombination layer was sputtered through a shadow mask at room temperature and power of 80 W. Then 10-nm NiO_x hole transport layer was deposited by RF magnetron sputtering at room temperature and power of 90 W using a NiO target, followed by annealing at 250 °C for 10 min in the air. The samples were then transferred to the glovebox. For the deposition of the self-assembly monolayers, Me-4PACz solution dissolved in ethanol (1 mg ml^{-1}) was spin coated onto the substrates at a speed of 3,000 rpm for 30 s, then annealed at 100 °C on a hot plate for 10 min. The perovskite absorber was formed at 3,500 rpm for 45 s using a solution of 1.4 M $Cs_{0.05}MA_{0.15}FA_{0.80}Pb(I_{0.75}Br_{0.25})_3$ in a DMF: DMSO (4:1) solvent mixture via spin coater (Brand-REESEEN, PvS-mini9, Jiangyin J. Wanjia Technology Co. Ltd). The ethyl acetate (350 μl) was dropped on the film at 25 s and post-annealed at 100 °C for 30 min in a glovebox. For the electron transport stack, 10 nm C_{60} was deposited by thermal evaporation, and 15 nm SnO_2 was deposited using ALD (E200SP, SuperALD). The tetrakis (dimethylamino) tin (IV) (TDMASn) purchased from Ai Mou Yuan Scientific Equipment (Nanjing) was used as the precursor of SnO_2. Next, a 40-nm-thick IZO layer was sputtered through a shadow mask at room temperature using 3-inch target (In_2O_3 with 10 wt% ZnO). Silver electrode (500 nm) was then thermally evaporated at a rate of 2.0 Å s^{-1} through a designed mask. Finally, 100-nm magnesium fluoride (MgF_x) as an antireflective layer was thermally evaporated at the rate of 1.0 Å s^{-1}.

Simulation

The PCE limits of different TOPCon architectures were simulated via Quokka 3. The key parameters are listed in Supplementary Table 1. The free energy loss analysis of the champion Finger TOPCon solar cell was performed via Quokka 3, and the key parameters are listed in Supplementary Table 2.

Characterizations

The illuminated *I*–*V* curves and EQE of the Finger TOPCon solar cells were tested and certified under standard conditions (25 °C, 1,000 W m^{-2}) by the Institute for Solar Energy Research in Hamelin. The relative humidity of the ambient during measurement is 30.5 ± 3.0%. One *I*–*V* curve consists of 165 non-uniform voltage steps. The data density is higher around voltage of zero, maximum power point and V_{OC} and at the very steep part of the *I*–*V* curve between maximum power point and maximum power point. The effective lifetimes, J_0 and iV_{OC} were measured by photoconductance decay measurement using a Sinton WCT-120 with the n-Si substrate. All the involved samples are symmetrically coated on both sides, such as hydrogenated SiO_x/poly-Si(n^+) or bilayer SiO_x/poly-Si(p^+) on both sides. The SiO_x thicknesses were first calibrated on monitor wafers using spectroscopic ellipsometry (SE-VE-L, Eoptics) and subsequently confirmed by cross-sectional TEM (FEI Titan Cubed Themis G2 300). After this initial calibration, routine process monitoring relied on ellipsometry-based thickness fitting. The Raman spectra were measured with a Horiba LabRAM Odyssey Raman spectrometer with a 325-nm excitation laser. Silicon layers were deposited on glass substrates using the same deposition parameters as the devices. The doping profiles were measured by ECV. The microstructure, selected area electron diffraction (SAED), HAADF and EDX mapping of SiO_x and poly-Si were investigated by high-resolution TEM (FEI Titan Cubed Themis G2 300). The morphology of texture, poly-Si, Ag electrode, Ag power, pitch and the width of the printed fingers were observed by SEM. The contact resistivity was measured by the transfer length method using photovoltaic tools (TLM SCAN). The stability of single-cell Finger TOPCon module was investigated under accelerated DH, LID test (at 50 °C under 1,000 W m^{-2} for 60 h per month) and LeTID degradation (324 h at 75°C under irradiance of 165–650 kW h m^{-2}). The UV stability was evaluated under 250 W m^{-2} by LED-based solar simulator for 240 h at 60 °C and 40% humidity. The illuminated *J*–*V* curves of the perovskite/TOPCon tandem solar cell were certified under standard testing conditions (25 °C, 1,000 W m^{-2}) by the National Center of Inspection on Solar Photovoltaic Products Quality. The voltage step was set to 0.02 V. The reflectance of light-trapping structures and EQE of the tandem device were measured by QE-R3018 system from EnliTech. The perovskite top cells and silicon bottom cells were measured independently by saturating with long-pass (850 nm) and low-pass (455 nm) filters, respectively. The operational stability of encapsulated perovskite/TOPCon tandem solar cells was carried out at the MPP with continuous AM 1.5 G illumination under LED-based solar simulator at room temperature in a nitrogen atmosphere using a home-made stability test system.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available within the Article and its Supplementary Information. Source data are provided with this paper.

References

1. Allen, T. G., Bullock, J., Yang, X., Javey, A. & De Wolf, S. Passivating contacts for crystalline silicon solar cells. *Nat. Energy* **4**, 914–928 (2019.
2. Gao, K. et al. Progress and future prospects of wide-bandgap metal-compound-based passivating contacts for silicon solar cells. *Adv. Mater.* **34**, 2200344 (2022).
3. Lin, H. et al. Silicon heterojunction solar cells with up to 26.81% efficiency achieved by electrically optimized nanocrystalline-silicon hole contact layers. *Nat. Energy* **8**, 789–799 (2023).
4. Yu, C. et al. Industrial-scale deposition of nanocrystalline silicon oxide for 26.4%-efficient silicon heterojunction solar cells with copper electrodes. *Nat. Energy* **8**, 1375–1385 (2023).
5. Yan, D. et al. Polysilicon passivated junctions: the next technology for silicon solar cells?. *Joule* **5**, 811–828 (2021).
6. Green, M. A. et al. Solar cell efficiency tables (version 65). *Prog. Photovolt. Res. Appl.* **33**, 3–15 (2024).
7. Green, M. A. et al. Solar cell efficiency tables (version 66). *Prog. Photovolt. Res. Appl.* **33**, 795–810 (2025).
8. Deng, S. et al. Mitigating parasitic absorption in poly-Si contacts for TOPCon solar cells: a comprehensive review. *Sol. Energy Mater. Sol. Cells* **267**, 112704 (2024).
9. Yang, Z. et al. Dual-side electrical refinement enables efficient industrial tunnel-oxide-passivating-contact silicon solar cells. *Nat. Energy* https://doi.org/10.1038/s41560-026-01982-2 (2025).
10. Li, R. et al. Optimizing performance and stability in textured 2 T perovskite/silicon tandem photovoltaic devices through self-assembled monolayer-mediated doping strategies. *Chem. Eng. J.* **518**, 164850 (2025).
11. Wang, Q. et al. 26.35%-efficiency and high-bifaciality n-TOPCon solar cells enabled by UV-ps laser-induced selective modification of double-layered SiO_x /n^+ -poly-Si passivating contacts. *Energy Environ. Sci.* **18**, 9217–9229 (2025).
12. Richter, A. et al. N-type si solar cells with passivating electron contact: identifying sources for efficiency limitations by wafer thickness and resistivity variation. *Sol. Energy Mater. Sol. Cells* **173**, 96–105 (2017).
13. Schmidt, J., Peibst, R. & Brendel, R. Surface passivation of crystalline silicon solar cells: present and future. *Sol. Energy Mater. Sol. Cells* **187**, 39–54 (2018).
14. Limodio, G. et al. Front and rear contact si solar cells combining high and low thermal budget si passivating contacts. *Sol. Energy Mater. Sol. Cells* **194**, 28–35 (2019).
15. Lozac'h, M. & Nunomura, S. Role of silicon surface, polished ⟨100⟩ and ⟨111⟩ or textured, on the efficiency of double-sided TOPCon solar cells. *Prog. Photovolt. Res. Appl.* **28**, 1001–1011 (2020).
16. Lozac'h, M. Double-sided TOPCon solar cells on textured wafer with ALD SiO_x layer. *Sol. Energy Mater. Sol. Cells* **207**, 110357 (2020).
17. Meyer, F. et al. Localisation of front side passivating contacts for direct metallisation of high-efficiency c-si solar cells. *Sol. Energy Mater. Sol. Cells* **235**, 111455 (2022).
18. Larionova, Y. et al. Ultra-thin poly-Si layers: passivation quality, utilization of charge carriers generated in the poly-Si and application on screen-printed double-side contacted polycrystalline Si on oxide cells. *Sol. RRL* **4**, 2000177 (2020).
19. Dasgupta, S. et al. Novel process for screen-printed selective area front polysilicon contacts for TOPCon cells using laser oxidation. *IEEE J. Photovolt.* **12**, 1282–1288 (2022).
20. Young, D. L. et al. Reactive ion etched, self-aligned, selective area poly-Si/SiO_2 passivated contacts. *Sol. Energy Mater. Sol. Cells* **217**, 110621 (2020).
21. Chen, K. et al. Self-aligned selective area front contacts on poly-Si/SiO_x passivating contact c-Si solar cells. *IEEE J. Photovolt.* **12**, 678–689 (2022).
22. Padhamnath, P. Design, development and analysis of large-area industrial silicon solar cells featuring a full area polysilicon based passivating contact on the rear and selective passivating contacts on the front. *Sol. Energy Mater. Sol. Cells* **256**, 112351 (2023).
23. Richter, A. et al. Design rules for high-efficiency both-sides-contacted silicon solar cells with balanced charge carrier transport and recombination losses. *Nat. Energy* **6**, 429–438 (2021).
24. Zhou, Z. et al. Optimizing strategy of bifacial TOPCon solar cells with front-side local passivation contact realized by numerical simulation. *Sol. Energy Mater. Sol. Cells* **278**, 113189 (2024).
25. Fell, A., Schön, J., Schubert, M. C. & Glunz, S. W. The concept of skins for silicon solar cell modeling. *Sol. Energy Mater. Sol. Cells* **173**, 128–133 (2017).
26. Basnet, R. et al. Current status and challenges for hole-selective poly-silicon based passivating contacts. *Appl. Phys. Rev.* **11**, 011311 (2024).
27. Glunz, S. W. et al. Silicon-based passivating contacts: the TOPCon route. *Prog. Photovolt. Res. Appl.* **31**, 341–359 (2023).
28. Niewelt, T. et al. Reassessment of the intrinsic bulk recombination in crystalline silicon. *Sol. Energy Mater. Sol. Cells* **235**, 111467 (2022).
29. Fell, A. et al. Radiative recombination in silicon photovoltaics: modeling the influence of charge carrier densities and photon recycling. *Sol. Energy Mater. Sol. Cells* **230**, 111198 (2021).
30. Zheng, P. et al. Polysilicon passivating contacts in mass production: the pursuit of higher efficiencies. *IEEE J. Photovolt.* **14**, 80–84 (2024).
31. Kale, A. S., Young, D. L., Agarwal, S. & Stradins, P. Modifications of textured silicon surface morphology and its effect on poly-Si/SiO_x contact passivation for silicon solar cells. *IEEE J. Photovolt.* **9**, 1513–1521 (2019).
32. Kale, A. S. et al. Effect of crystallographic orientation and nanoscale surface morphology on poly-Si/SiO_x contacts for silicon solar cells. *ACS Appl. Mater. Interfaces* **11**, 42021–42031 (2019).
33. Guo, C. Influence of backside surface morphology on passivation and contact characteristics of TOPCON solar cells. *Sol. Energy* **258**, 278–288 (2023).
34. David, T. Stress, strain, and raman spectroscopy. *Spectroscopy* **34**, 10–21 (2019).
35. Moldovan, A. et al. Tunnel oxide passivated carrier-selective contacts based on ultra-thin SiO_2 layers. *Sol. Energy Mater. Sol. Cells* **142**, 123–127 (2015).
36. Stuckelberger, J. et al. Pre-annealing for improved LPCVD deposited boron-doped poly-Si hole-selective contacts. *Sol. Energy Mater. Sol. Cells* **251**, 112123 (2023).
37. Yang, G. et al. Will SiO_x -pinholes for SiO_x/poly-Si passivating contact enhance the passivation quality?. *Sol. Energy Mater. Sol. Cells* **252**, 112200 (2023).
38. Du, H. et al. Blistering-free carbon-doped polysilicon (n^+) passivating contact with high surface passivation properties prepared by industrial tube PECVD. *Mater. Sci. Semicond. Process.* **170**, 107969 (2024).
39. Cheng, H. et al. Emitter formation with boron diffusion from PECVD deposited boron-doped silicon oxide for high-efficiency TOPCon solar cells. *Sol. Energy Mater. Sol. Cells* **240**, 111713 (2022).

40. Padhamnath, P. et al. Progress with passivation and screen-printed metallization of boron-doped monoPoly™ layers. *Sol. Energy* **231**, 8–26 (2022).
41. Yu, X. et al. Effect of silver powder microstructure on the performance of silver powder and front-side solar silver paste. *Materials* **17**, 445 (2024).
42. Hilali, M. M., Pal, S., More, R. V., Saive, R. & Ardekani, A. M. Sheared thick-film electrode materials containing silver powders with nanoscale surface asperities improve solar cell performance. *Adv. Energy Sustain. Res.* **3**, 2100145 (2022).
43. Glatthaar, R. et al. Contact formation of silver paste and atmospheric pressure chemical vapor deposition (n) poly-silicon passivating contacts on planar and textured surfaces. *Phys. Status Solidi A* **219**, 2200501 (2022).
44. Peibst, R. et al. On the chances and challenges of combining electron-collecting *n* POLO and hole-collecting al- p^+ contacts in highly efficient *p* -type c-Si solar cells. *Prog. Photovolt. Res. Appl.* **31**, 327–340 (2023).
45. Chen, X. et al. Biased plasma treated nickel oxide for high-efficiency perovskite/silicon tandem solar cells. *Adv. Mater.* **37**, 2504581 (2025).
46. Chen, Y. et al. Nuclei engineering for even halide distribution in stable perovskite/silicon tandem solar cells. *Science* **385**, 554–560 (2024).
47. Yang, L. et al. Modulating binding strength and acidity of benzene-derivative ligands enables efficient and hysteresis-free perovskite/silicon tandem solar cells. *Angew. Chem. Int. Ed.* **64**, e202500350 (2025).
48. Green, M. A. et al. Solar cell efficiency tables (version 58). *Prog. Photovolt. Res. Appl.* **29**, 657–667 (2021).
49. Green, M. A. et al. Solar cell efficiency tables (version 60). *Prog. Photovolt. Res. Appl.* **30**, 687–701 (2022).
50. Feldmann, F., Bivour, M., Reichel, C., Hermle, M. & Glunz, S. W. Passivated rear contacts for high-efficiency n-type si solar cells providing high interface passivation quality and excellent transport characteristics. *Sol. Energy Mater. Sol. Cells* **120**, 270–274 (2014).
51. Feldmann, F. et al. Tunnel oxide passivated contacts as an alternative to partial rear contacts. *Sol. Energy Mater. Sol. Cells* **131**, 46–50 (2014).
52. Green, M. A., Emery, K., Hishikawa, Y., Warta, W. & Dunlop, E. D. Solar cell efficiency tables (version 47). *Prog. Photovolt. Res. Appl.* **24**, 3–11 (2016).
53. Green, M. A. et al. Solar cell efficiency tables (version 50). *Prog. Photovolt. Res. Appl.* **25**, 668–676 (2017).
54. Chen, D. et al. 24.58% total area efficiency of screen-printed, large area industrial silicon solar cells with the tunnel oxide passivated contacts (i-TOPCon) design. *Sol. Energy Mater. Sol. Cells* **206**, 110258 (2020).
55. Green, M. A. et al. Solar cell efficiency tables (version 64). *Prog. Photovolt. Res. Appl.* **32**, 425–441 (2024).

Acknowledgements

X.Y. acknowledges the financial support from the National Key R&D programme of China (number 2022YFB4200203), the National Natural Science Foundation of China (numbers 62174114 and U24A2063), the Key Project of Jiangsu Provincial Basic Research Plan (number BK20243031) and the Department of Science and Technology of Jiangsu Province (BE2022023 and BE2022027). This work was supported by the 'Pioneer' and 'Leading Goose' R&D Program of Zhejiang Province (grant number 2025C01154). L.Y. acknowledges financial support from the Postdoctoral Fellowship Program of CPSF (number GZC20252252) and the China Postdoctoral Science Foundation (number 2025M770174).

Author contributions

X.Y. conceived the idea, designed the experiments and led the project. K.G. and J.M. fabricated the devices, performed the device characterizations and wrote the paper. Z.Y. and D.M. performed the Quokka simulations. L.Y., W.S., C.W., B.Y. and B.G. helped characterize and analyse the microstructure, optical and electrical properties of the poly-Si layer and TOPCon solar cells. P.Z., J.Y. and Xinyu Zhang helped with the TOPCon device optimization and analysis. S.W., F.C., J.W., M.X. and C.L. helped with the fabrication and characterization of the tandem devices. S.W. and W.L. performed the stability measurements of the tandem device. X.Y. and Xiaohong Zhang supervised the project. All authors contributed to the discussion of the results and revision of the manuscript.

Competing interests

J.M., J.W., P.Z., M.X., J.Y. and Xinyu Zhang are employees of Zhejiang Jinko Solar Co. Ltd. The remaining authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41560-026-02007-8.

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Peer review information *Nature Energy* thanks Pingqi Gao, Franz-Josef Haug and Armin Richter for their contribution to the peer review of this work.

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Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	335.5 cm2 active area for Si cell; 0.9971 cm2 aperture area for tandem cell. *Explain why this information is not reported/not relevant.*
Method used to determine the device area	☒ Yes ☐ No	The area of Si cells was measured using an optical scanning system. The area of tandem solar cell is determined by mask, whose area was measured using an optical scanning system. *Explain why this information is not reported/not relevant.*

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	J-V curves of tandem cell in Fig. 5b is in both direction in the main text. Forward direction is enough for Si cells.
Voltage scan conditions	☒ Yes ☐ No	Forward direction for Si cell; 165 points; integral time 5 ms; delay time 5ms.The voltage scan for tandem cell were done both Jsc to Voc and Voc to Jsc under the scan speed of 200 mV/s and the dwell time was not applied. *Explain why this information is not reported/not relevant.*
Test environment	☒ Yes ☐ No	"Method" section in the main text. *Explain why this information is not reported/not relevant.*
Protocol for preconditioning of the device before its characterization	☒ Yes ☐ No	No light soaking or electric-poling preconditioning. *Explain why this information is not reported/not relevant.*
Stability of the J-V characteristic	☒ Yes ☐ No	Fig. 5c in the main text. *Explain why this information is not reported/not relevant.*

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes ☐ No	Discussion in "Perovskite/TOPCon tandem solar cell" section in the main text. *Explain why this information is not reported/not relevant.*
Related experimental data	☒ Yes ☐ No	Relevant statement on Fig. 5b in the main text. *Explain why this information is not reported/not relevant.*

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	Fig. 2e & Fig. 5d. *Explain why this information is not reported/not relevant.*
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☐ Yes ☒ No	*State where this information can be found in the text.* Certification reports do not contain this comparison.

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☒ Yes ☐ No	"Method" section in the main text. *Explain why this information is not reported/not relevant.*

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	"Method" section in the main text. *Explain why this information is not reported/not relevant.*
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The spectrum in ISFH is tested before each measurement of the calibration object using a spectroradiometer. The spectroradiometer is traceable to reference lamps which were calibrated at the Physikalisch-Technische undesanstalt PTB (Braunschweig). The reference cell used in CPVT is unknown. *Explain why this information is not reported/not relevant.*
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	Spectral mismatch factor MM for Si cells is 1.002. Certification reports do not contain this information for tandem cells. *Explain why this information is not reported/not relevant.*

6. Mask/aperture

Size of the mask/aperture used during testing	☒ Yes ☐ No	1 cm2 for tandem cell; total area for Si cell. *Explain why this information is not reported/not relevant.*
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	*Report the difference in the short-circuit current density values measured with the mask and aperture area.* Not relevant for Si cell. The results for tandem are always with aperture.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	ISFH for Si cell; CPVT for tandem cell. *Explain why this information is not reported/not relevant.*
A copy of any certificate(s)	☐ Yes ☒ No	*Certificate copies should be provided in the Supplementary information. Please state the supplementary item number.* Fig. 2d for Si solar cell in the main text. Fig. 5b in the main text & Fig. S13 for tandem cell in the supplementary Information.

8. Statistics

Number of solar cells tested	☒ Yes ☐ No	100 devices for Si solar cell and tandem cell. *Explain why this information is not reported/not relevant.*
Statistical analysis of the device performance	☒ Yes ☐ No	Fig. 2d & Fig. 5b and corresponding discussion in the main text. *Explain why this information is not reported/not relevant.*

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Fig. 2h-j & Fig. 5f, corresponding figure captions and "Method" in the main text. *Explain why this information is not reported/not relevant.*